Radial Dimensionality Theory IV: Tidal Deformability Ceilings under Solar-Constrained Density-Dependent Dimensionality

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We recompute conditional tidal deformabilities on the full-density-law neutron-star backgrounds of Paper III using an explicitly prescribed GR-form quadrupole equation. The same CompOSE RG(SLy4) EOS, finite-pressure surface and updated solar model/proxy region are used in both papers. At the selected solar-amplitude peak and $1.4M_\odot$, GR gives $\Lambda = 298.0203$ and Embedding 1 gives $\Lambda = 286.07414$, a -4.00852% change. The exact radius-only change is -2.10732% and the Love number changes -1.94212%; the full response is integrated, not inferred from radius scaling. Embedding 2 gives -3.78878% under the same additional perturbation prescription. Sixty augmented stellar outputs, 21 paired comparisons and 24 controls are retained. Tightening and a linear-metric perturbation formulation change selected fractional tidal shifts by less than 2.32×10^{-9}. The calculation uses a prescribed Schwarzschild exterior; the interior metric gradient differs from it by about 1.29793% at the primary finite cutoff. The results therefore depend on an unresolved boundary prescription. They remain empirically testable conditional samples, not global, EOS-independent or covariantly completed predictions.

VERSION NOTE: V7

This version retains the published title, but the word Ceilings no longer describes an established result: the revision reports conditional tidal samples, not global upper bounds. It integrates the full Love-number and radius response using the specified stellar backgrounds, GR-form perturbation equation and Schwarzschild exterior extraction. That exterior is a prescribed part of the calculation, not a matching result derived from the interior; the metric-gradient mismatch at the finite cutoff is about 1.30 percent in the primary 1.4-solar-mass example. This mismatch is a matching diagnostic, not an error bar on the tidal deformability.

I. SCOPE AND COMMON INPUTS

Tidal response probes the stellar radius and internal mass distribution through $\Lambda = (2/3)k_2 C^{-5}$, with compactness $C = GM/(Rc^2)$ [1, 2]. To compare the RDT background prescriptions, we integrate their perturbations under specified additional assumptions. No new observational likelihood or population-detectability analysis is performed here.

Paper I's source-precision diagonal proxy region is conditional on its solar polytrope, error model and helioseismic proxy [3, 4]. Its nominal $\Delta Q = 2.71$ slice endpoint is denoted $A_{\rm lim}(\alpha)$; coverage has not been calibrated. The locally refined solar-amplitude peak used for the main examples is

$$(\alpha_\star, A_\star) = (0.1244971139, 0.0087502019). \tag{1}$$

It is not established as the maximum tidal-response point. The shared law is $\delta = -3Ax/(1+x)$, $x = (\rho_b/\rho_{\rm ref})^\alpha$, with physical baryon density and $\rho_{\rm ref} = 150\ \mathrm{g\,cm^{-3}}$. Its high-density asymptote at this point is $d_{\rm eff} = 2.973749394$, not a strict universal lower bound above 2.974. The full density dependence is retained, including at small α.

II. BACKGROUND AND PERTURBATION PRESCRIPTIONS

A. Checked stellar backgrounds

We use Paper III's two explicit pressure-flux prescriptions [4, 5]. In geometrized units $G = c = 1$, let $h = (r/r_\star)^\delta$, $f = 1 - 2m/r$ and $q = B_\delta/(hB_0)$, where m is ordinary energy mass, B_0 and B_δ are the ordinary and weighted baryon integrals, and E_δ is the weighted energy integral. The density law is evaluated with the physical CGS baryon density before consistent conversion of sources. Then

$$p' = -(e+p)g, \qquad m' = 4\pi r^2 e, \tag{2}$$

$$g = \frac{D}{r^2 f}, \qquad D_1 = q(m + 4\pi r^3 p), \tag{3}$$

$$D_2 = E_\delta/h + 4\pi r^3 p. \tag{4}$$

Both use ordinary mass in f and reduce to GR through the same equations at $A = 0$. The primary physical reference length is 10 km; it is not the Lane–Emden reference of Papers I–II.

The source-checked 197-point CompOSE RG(SLy4) table is the only EOS used [4, 6]. Energy density and baryon rest-mass density are distinct. The surface is at $P_s = 1.001 P_{\rm min}$, without extrapolation. Corrected central pressure and weighted-integral seeds are inherited from Paper III. Rooted target masses and augmented background radii agree with the checked five-state solver within the stated tolerances. APR remains withdrawn;

neither the EOS interpolation tests nor the Douchin–Haensel source regression constitutes a multi-EOS uncertainty analysis.

B. GR-form quadrupole equation

For a static even-parity $\ell = 2$ perturbation, we prescribe the GR-form equation for $y = rH'/H$ [1, 7]:

$$ry' + y^2 + Fy + r^2Q = 0, \tag{5}$$

$$F = \frac{1 + 4\pi r^2(p - e)}{f}, \tag{6}$$

$$Q = \frac{4\pi}{f}[5e + 9p + (e + p)de/dp] - \frac{6}{fr^2} - (\nu')^2. \tag{7}$$

The chosen metric gradient is $\nu' = 2g$, preserving the stated hydrostatic relation. This does not derive Eq. (5) from modified covariant field equations. Retaining the same F, the energy-derivative term and ordinary mass in f is an additional prescription for both embeddings. Extra gravitational perturbation degrees of freedom are not evolved.

The regular seed is $y = 2 + y_2 r_0^2$, where

$$y_2 = -\frac{4\pi}{7}\left[\frac{e_c}{3} + 11p_c + (e_c + p_c)\left(\frac{de}{dp}\right)_c\right]. \tag{8}$$

A separate integration evolves H/r^2 and its logarithmic derivative, then reconstructs y. This checks the numerical formulation of the same prescribed equation; it is not a second gravity theory.

C. Finite-cutoff extraction and exterior matching

The primary surface convention is

$$y_{\rm out} = y_{\rm in} - \frac{4\pi R^3 e_s}{m + 4\pi R^3 p_s}. \tag{9}$$

The pressure term is present in the GR density-discontinuity relation [8]. Applying that relation at a truncated modified background is an extraction prescription, not a derived physical shell. Energy is evaluated at the actual cutoff. The corrected Schwarzschild $k_2(C, y_{\rm out})$ formula [1] then defines the reported Λ. Decimal arithmetic in this final expression avoids cancellation in the low-compactness benchmarks.

The exterior is prescribed, not established by matching. The tidal calculation uses a Schwarzschild exterior. The computed interior does not yet satisfy smooth metric-gradient matching to that exterior at the chosen surface, so the reported tidal response depends on this unresolved boundary prescription. Specifically,

$$\eta = \frac{\nu'_{\rm in}}{2m/[R^2(1 - 2m/R)]} - 1 \tag{10}$$

TABLE I. Common-point background/tidal record shared with Paper III. Masses are EOS-domain peaks, in solar masses; radii are at $1.4M_\odot$.

Emb.	$M_{\rm dom}/M_\odot$	$R_{1.4}$ (km)	$\Lambda_{1.4}$
1	2.0352165	11.636192	286.0741
2	2.0380823	11.638859	286.7290

is 1.29793% (Embedding 1) and 1.31654% (Embedding 2) at Eq. (1) and $1.4M_\odot$. An additive normalization of ν cannot remove this derivative mismatch. Density-law vacuum nulling alone does not identify the surviving integrated field charge with ordinary energy mass. No transition layer, shell stress or alternative exterior is derived here. The mismatch is *not* an estimated error bar on Λ.

We also retain zero-pressure GR, no-jump and frozen-hydrostatic-gradient extractions from each same interior. Their largest span in the primary paired fractional tidal shifts is 2.32×10^{-8}. This small cutoff sensitivity does not bound the unresolved exterior theory uncertainty. A specified phenomenological prescription can still be tested directly against data; outside endorsement is not a prerequisite.

III. NUMERICAL VERIFICATION

The primary calculation resolves every EOS pressure knot in $t = \ln(P_c/P)$ and uses the analytic derivative of the selected log-linear energy interpolant. It does not extrapolate or use a derivative stencil outside the table. DOP853 settings are rtol 2×10^{-9}, background atol 10^{-17}, tidal atol 2×10^{-11}, maximum step .08 and central radius 100 cm. Fixed masses are rooted with the checked background; augmented mass and radius must agree fractionally within 5×10^{-7}, and target masses within $5 \times 10^{-7} M_\odot$. Failed integrations and invalid inputs stop admission rather than being silently skipped.

The Newtonian $n = 1$ polytrope, extrapolated from compactness .002, .004 and .008, gives $k_2(0) =$ .2599086282 against the analytic .2599088773. An incompressible extraction check tends to .75. The GR $1.4M_\odot$ result is $R = 11.685866$ km, $k_2 = 0.07746217$ and $\Lambda = 298.0203$. Both embeddings at $A = 0$ and the linear-H form pass the fixed relative 10^{-6} tidal gate. Invalid-input and injected-failure tests also pass.

The retained dataset contains 60 augmented stellar outputs, 21 paired comparisons and 24 controls, plus background-root evaluations. Three selected pairs (the solar peak in both embeddings and low alpha in Embedding 1) each undergo eight controls. Maximum changes in fractional Λ shift are:

TABLE II. Computed conditional tides at the common solar-amplitude peak and 10 km reference length. Shifts are percent. Both embeddings use the same EOS, finite cutoff and prescribed perturbation/exterior extraction.

$M/M_\odot$	$\Lambda_{\rm GR}$	Λ_1	$\Delta\Lambda_1$ (%)	Λ_2	$\Delta\Lambda_2$ (%)
1.20	790.8891	761.6252	-3.70013	762.8615	-3.54381
1.27	559.0919	537.8980	-3.79076	538.8793	-3.61525
1.40	298.0203	286.0741	-4.00852	286.7290	-3.78878
1.46	223.6417	214.3848	-4.13914	214.9337	-3.89373
1.60	114.0457	108.8484	-4.55718	109.2203	-4.23106

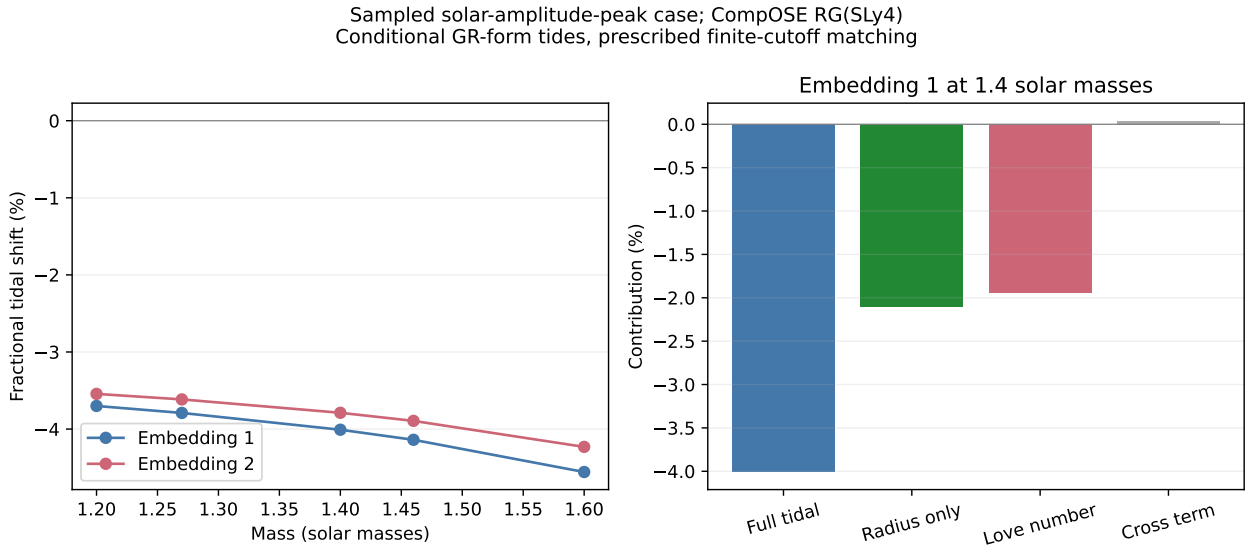


FIG. 1. Left: sampled fractional tidal shifts at the common solar point; lines connect samples rather than define a bound. Right: Embedding 1 at $1.4M_\odot$, separating the full tidal response, exact radius-only change, Love-number change and multiplicative cross term. The binary combination is reported separately.

Control	Maximum change
Tight integration	2.314×10^{-9}
Linear-H, tight integration	2.314×10^{-9}
Start radius 10 / 1,000 cm	$(2.928, 3.080) \times 10^{-9}$
PCHIP EOS interpolation	1.616×10^{-5}
Surface at $10/100P_{\rm min}$	$(1.482, 4.092) \times 10^{-5}$
Unmodified Einstein ν' diagnostic	1.342×10^{-3}

Tight settings use rtol 2×10^{-11}, background atol 10^{-19} and maximum step .03. The first two controls meet the predefined 10^{-6} paired admission threshold. EOS/surface changes are separate scope sensitivities. The last diagnostic deliberately changes the metric coefficient in the tidal operator while keeping the background fixed; it breaks the chosen hydrostatic relation and is not a self-consistent alternative theory or a theory-error estimate.

IV. RESULTS

A. Sampled tidal response

The five main masses at Eq. (1) are shown in Table II. Other pairs sample exponents .02, .5 and the Paper III NS-extremum slice, reference lengths 5/20 km and the historical (.0087,.122) point. No tidal-domain extremum or continuous envelope is certified.

At $1.4M_\odot$, Embedding 1 changes Λ by -4.00852%. Its radius changes -0.42506% and its Love number changes -1.94212%. The exact identity is

$$\frac{\Lambda_{\rm mod}}{\Lambda_{\rm GR}} = \frac{k_{2,\rm mod}}{k_{2,\rm GR}} \left(\frac{R_{\rm mod}}{R_{\rm GR}}\right)^5 \left(\frac{M_{\rm GR}}{M_{\rm mod}}\right)^5 . \qquad (11)$$

The tiny actual mass-root residual is retained. The radius-only tidal change is -2.10732%, and the radius–Love cross term is $+0.04093$ percentage points. Thus $5\Delta R/R + \Delta k_2/k_2$ is only a first-order approximation. Embedding 2 gives $\Lambda_{1.4} = 286.72902$ and -3.78878%; the difference between embeddings is not an exhaustive uncertainty bound.

B. Shared background and binary example

Table I repeats the exact common-point record used in Paper III. The background domain mass is not a true modified maximum mass: the sequence reaches the upper EOS edge. Its appearance beside a tidal value does not remove the exterior assumptions stated above.

For component masses $m_1 = 1.46M_\odot$ and $m_2 = 1.27M_\odot$, the usual mass-weighted combination [2] is

$$\widetilde{\Lambda} = \frac{16}{13}\frac{(m_1 + 12m_2)m_1^4\Lambda_1 + (m_2 + 12m_1)m_2^4\Lambda_2}{(m_1 + m_2)^5}. \quad (12)$$

It gives GR 359.54855, Embedding 1 345.43908 and Embedding 2 346.16640. These replace the rounded 359/345 example. They are not a new GW170817 likelihood analysis, a detection or a claim that this model removes EOS/measurement degeneracies. Observational tests must specify their mass, EOS, waveform and error treatment. The binary observable is not plotted as though it were a single-star $\Lambda(M)$.

V. INTERPRETATION AND REVISION HISTORY

The unchanged V6 pipeline at its historical (.0087,.122) point gives a 3.98336% reduction at $1.4M_\odot$. The checked solver at those same parameters gives 3.97822%; the updated solar point gives 4.00852%. This separates numerical changes from input changes. V6 used an uncorrected pressure seed, a finite-difference energy derivative, surface energy above the cutoff and a zero-pressure jump convention. Its plotting loop could skip failed stars. Those historical files remain preserved; they do not supply the current verification evidence.

Earlier force-multiplier enhancements remain results of a different prescription. The earlier EOS reconstruction error, unsupported APR predictions and mislabeled Newtonian validation remain withdrawn. Shared contraction signs are not a criterion for physical validity, and the current sampled reductions are not universal ceilings.

The calculation now establishes reproducible, numerically controlled outputs of an explicitly stated conditional tidal prescription. It does not establish a covariant perturbation sector, a physical surface layer or smooth matching to the prescribed Schwarzschild exterior. These are specific unresolved model ingredients, not a requirement for external endorsement before empirical comparison.

CODE AND REVISION AVAILABILITY

Producing scripts, inputs, saved results and numerical audit evidence are publicly available in the shared RDT reproducibility archive. The accompanying source package, `merrill2025_tidal_deformability_code`, includes the manuscript, dependencies and resource-controlled reproduction commands. Its `reproduction/CAPSULE.json` records file hashes; `reproduce.py` verifies the package and rebuilds the paper from retained scientific outputs. Full scientific runs are separate from this saved-result reproduction. Historical scripts and results remain separately identified.

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